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Trade: Artificial Intelligence (Machine Learning; Deep Learning; Communication)

Test Date & Time: 22/05/2026 02:00 PM

Test Center Name: Institute of Business Administration
Karachi

Examiner Name: Farhan /

Signatures

Candidate Signature

Examiner Signature & Stamp

Principal Signature

K.M Zarar

Instructions:

Please read instructions carefully before attempting paper:

- 1) All questions are compulsory.
- 2) The question paper consists of 21 questions divided into two sections A & B. Section A comprise of 20 MCQs of 1 marks each. Section B comprises of 1 practical task having 80 marks.
- 3) Mobile Phone not allowed.
- 4) Use separate paper for rough work.

Section A - Theory (Total Marks: 20, Time: 30 Minutes)

1- Seaborn is a Python data visualization library based on which other library?

- (a) Scikit-learn
- (b) Matplotlib
- (c) Pandas
- (d) Numpy

2- In Python, which function is used to apply a function to all elements in an iterable?

- (a) apply()
- (b) filter()
- (c) iterate()
- (d) map()

3- In univariate linear regression with gradient descent, what is the role of the gradient descent algorithm?

- (a) To calculate the slope of the regression line
- (b) To maximize the cost function and find optimal parameters
- (c) To minimize the cost function and find optimal parameters
- (d) To calculate the intercept of the regression line

4- In Linux, which command is used to update the package list?

- (a) apt-get
- (b) apt-upgrade
- (c) apt-list
- (d) apt-update

5- What is the primary application of natural language processing (NLP) solutions?

- (a) Image analysis
- (b) Video compression
- (c) Text analysis and understanding
- (d) Audio processing

6- Which type of data includes non-numerical information such as labels or categories?

- (a) Numerical data
- (b) Qualitative data
- (c) Categorical data
- (d) Quantitative data

7- What is the purpose of using keyword arguments in function calls?

- (a) To increase the performance of a function
- (b) To assign a name to an argument
- (c) To change the order of arguments
- (d) To make a function private

8- In Microsoft Azure, what networking solution allows you to isolate and secure your cloud resources by creating private network connections and controlling traffic flow between virtual machines and services?

- (a) Azure Active Directory
- (b) Azure Firewall
- (c) Azure Virtual Networks
- (d) Azure Security Center

9- Which data structure is commonly used in Pandas to store tabular data?

- (a) Arrays
- (b) Lists
- (c) DataFrames
- (d) Sets

10- In supervised machine learning, which component represents the function that the model is trying to learn and approximate?

- (a) Optimizer
- (b) Hypothesis
- (c) Cost function
- (d) Labeled data

11- In Pandas, which data structure is used to store one-dimensional labeled data, similar to a column in a spreadsheet?

- (a) List
- (b) DataFrame
- (c) Array
- (d) Series

12- Which symbol is used in a regular expression to match any single character?

- (a) ^
- (b) *
- (c) .
- (d) ?

13- What is dynamic polymorphism in object-oriented programming?

- (a) A technique to combine multiple classes into one
- (b) The ability of a subclass to override methods of its base class
- (c) A way to prevent inheritance between classes
- (d) The process of defining new data types

14- In statistics, which term describes a variable that can take on distinct, separate values?

- (a) Quantitative
- (b) Qualitative
- (c) Discrete
- (d) Continuous

15- In deep learning, which type of neural network is commonly used for image classification tasks, leveraging 2D convolutional layers to automatically learn hierarchical features from images?

- (a) 2D Convolutional Neural Network (CNN)
- (b) Autoencoder
- (c) Multi-Layer Perceptron (MLP)
- (d) Recurrent Neural Network (RNN)

16- What does broadcasting in Numpy allow you to do?

- (a) Convert arrays to different data types
- (b) Perform element-wise operations on arrays with different shapes
- (c) Combine arrays into a single array
- (d) Resize arrays to match the largest array

17- What is an object in object-oriented programming?

- (a) A block of code that is executed once
- (b) A function that defines a class
- (c) An instance of a class with its own properties and methods
- (d) A value that can be changed during runtime

18- What are the fundamental building blocks of programming that manipulate data?

- (a) Operators
- (b) Keywords
- (c) Variables
- (d) Functions

19- What is the purpose of list comprehension in Python?

- (a) To create complex nested lists
- (b) To manipulate and filter lists using a concise syntax
- (c) To write long and verbose loops
- (d) To generate random numbers

20- Which command is used to display the contents of a file in a Unix-like operating system?

- (a) read
- (b) view
- (c) cat
- (d) open

__The End__